

Linear Block Codes

Syllabus

Linear block codes : Coding, Syndrome and error detection, Error detection and correction capability, Standard array and syndrome decoding.

Chapter Contents

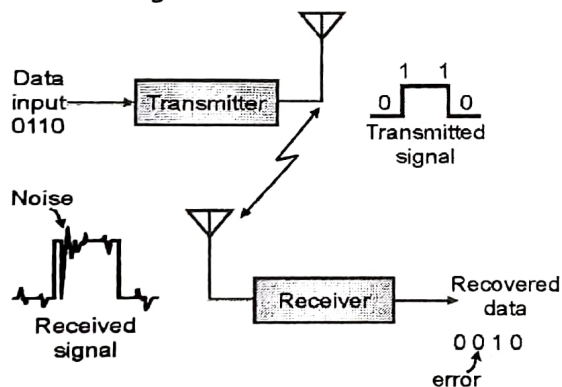
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8.1 Introduction :

- The desirable characteristics of a digital communication system are high data rate, minimum signal power, highly reliable transmission of data and minimum bandwidth requirement.
- However practical communication channels are noisy and have a smaller bandwidth.
- If we try to minimize the transmitted power, then the signal to noise ratio will reduce.
- This increases the error probability P_e and reduces the system reliability.
- In digital communication, errors are introduced during the transmission of data from the transmitter to receiver due to noise or some other reasons.

Need of error control coding :

- When transmission of digital signals takes place between two systems such as computers, the signal get contaminated due to the addition of "Noise" to it.
- The noise can introduce an error in the binary bits travelling from one system to the other.
- That means a 0 may change to 1 or a 1 may change to 0 as shown in Fig. 8.1.1.



(E-2064)Fig. 8.1.1 : Need of error control coding.

- These error can become a serious threat to the accuracy of the digital system.
- Therefore it is necessary to detect and correct the errors.
- In data communication, errors are introduced during the transmission of data from the transmitter to receiver due to noise or some other reasons.
- The reliability of data transmission will be severely affected due to these errors.
- In order to improve the reliability of data transmission, the designer will have to increase the signal power or

reduce the noise spectral density N_0 so as to maximize the ratio E_b / N_0 .

- But practically there is a limitation on the maximum value of the ratio E_b / N_0 . We cannot increase the ratio beyond this limit.
- Hence for a fixed value of E_b / N_0 , we have to use some kind of "coding" in order to improve the quality of the transmitted signal.
- Another advantage of using coding is that we can reduce the required value of E_b / N_0 if the error rate is predefined and remains fixed at that value.
- This will in turn reduce the required transmitted power and the size of antenna.

How to detect and correct errors ?

- For the detection, and / or correction of these errors, one or more than one extra bits are added to the data bits at the time of transmitting.
- These extra bits are called as **parity bits**. They allow the detection or sometimes correction of the errors. However they do not contain any information. Hence they are also called as the redundant bits.
- The data bits along with the parity bits form a code word.
- In other words, the error correcting codes are produced by adding redundancy to the original message before its transmission.

8.1.1 Error Control Techniques :

- The error control techniques can be divided into two types :
 1. Error detection techniques
 2. Error correction techniques
- The error detecting techniques are capable of only detecting the errors. They cannot correct the errors.
- The error correcting techniques are capable of detecting as well as correcting the errors.

8.1.2 Encoding and Decoding :

Encoding :

- The process of adding the parity bits to the data bits at the time of transmission is known as encoding.
- These **parity** bits or **redundant** bits do not contain any information.
- However, they allow the detection or sometimes correction of the errors.

The data bits along with the parity bits form a code word as shown in Fig. 8.1.2.

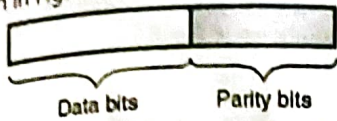


Fig. 8.1.2 : Structure of a transmitted code word

produced at the output of the encoder as shown in Fig. 8.1.3.

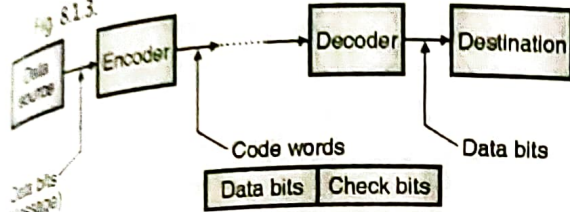


Fig. 8.1.3 : Encoding and decoding

Decoding :

The process of obtaining the message bits from the received code word at the receiver is called as decoding.

Block diagram :

- Fig. 8.1.3 shows the encoding and decoding processes.
- The source generates the data (message) in the form of binary symbols.
- The encoder accepts these bits and adds the check (parity) bits to them to produce the code words.
- These code words are transmitted towards the receiver.
- The decoder separates out the data and check bits.
- It uses the parity bits to detect and correct errors if they are present in the received code words.
- The data bits are then passed on to the destination.

Redundancy :

- Redundancy involves transmission of extra parity bits or check bits along with the data bits.
- Higher the number of parity bits, higher the redundancy.

8.2 Definitions Related to Codes :

8.2.1 Code Word :

- The code word is the n bit encoded block of bits.
- As already seen it contains k message bits and $(n-k)$ parity or redundant bits, as shown in Fig. 8.2.1.

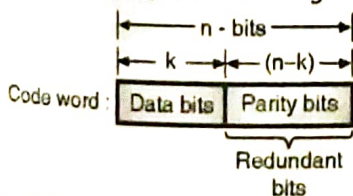


Fig. 8.2.1 : Structure of a transmitted code word

8.2.2 Code Rate :

SPPU : Dec. 19

University Questions

- Q. 1 Define and give example : Code rate
(Dec. 19, 2 Marks)

Definition :

- The code rate is defined as the ratio of the number of message bits (k) to the total number of bits (n) in a code word.

$$\therefore \text{Code rate } (r) = \frac{k}{n} \quad \dots(8.2.1)$$

Example :

- The code rate of a (7,4) code is $r = 4/7$.

8.2.3 Hamming Weight $w(x)$:

SPPU : March 19, Dec. 19

University Questions

- Q. 1 Explain following terms with reference to Linear Block Code Hamming weight. (March 19, 2 Marks)
- Q. 2 Define and give example Hamming weight
(Dec. 19, 2 Marks)

Definition :

- The Hamming weight of a code word x is defined as the number of non-zero elements in the code word.
- Hamming weight of a code vector (code word) is the distance between that code word and an all zero code vector. (a code having all elements equal to zero).
- For example, the Hamming weight of the following code word is 3, as it contains 3 non-zero elements.

$$\text{Code word} = 1001001$$

8.2.4 Code Efficiency :

Definition :

- The code efficiency is defined as the ratio of message bits to the number of transmitted bits per block.

$$\therefore \text{Code efficiency} = \text{Code rate} = \frac{k}{n} \quad \dots(8.2.2)$$

8.2.5 Hamming Distance :

SPPU : May 17, March 19, Dec. 19

University Questions

- Q. 1 Explain the following : Hamming distance.
(May 17, 2 Marks)
- Q. 2 Explain following terms with reference to Linear Block Code Hamming distance.
(March 19, 2 Marks)
- Q. 3 Define and give example : Hamming distance
(Dec. 19, 2 Marks)

**Definition :**

- Consider two code vectors (or code words) having the same number of elements.
- The "Hamming distance" or simply distance between the two code words is defined as the number of locations in which their respective elements differ.

Example :

- For example consider the two code words given below :

Codeword - 1 :

1	1	1	1	0	1	0	0
---	---	---	---	---	---	---	---

Codeword - 2 :

0	1	0	1	1	1	1	0
---	---	---	---	---	---	---	---

(E-1705)

- Note that the bits 2, 4, 7 and 8 are different from each other. Hence Hamming distance is 4.

Ex. 8.2.1 : Find the Hamming weight of the following code vector : $x = 1\ 1\ 0\ 1\ 0\ 1\ 0\ 0$

Soln. :

- As the number of non-zero elements in the above code word is 4, the Hamming weight $W(x) = 4$.

8.2.6 Minimum Distance $d_{\min}$:**SPPU : May 07, Dec. 19****University Questions**

- Q. 1** Define term related to convolution codes - minimum distance. (May 07, 2 Marks)
- Q. 2** Define and give example : Minimum hamming distance. (Dec. 19, 2 Marks)

Definition :

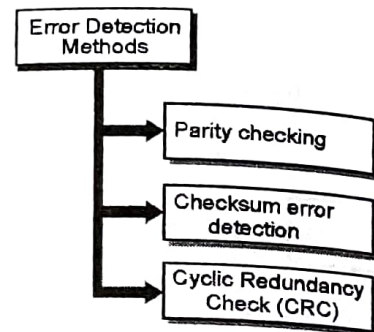
- The minimum distance " $d_{\min}$ " of a linear block code is defined as the smallest Hamming distance between any pair of code vectors in the code.
- Therefore the minimum distance is same as the smallest Hamming weight of difference between any pair of code vectors.
- It can be proved that the minimum distance of a linear block code is the smallest Hamming weight of the non-zero code vectors in the code.

8.3 Error Detection Techniques :

- A number of methods (techniques) are available for the detection and correction of errors introduced in the transmitted signal.
- When a codeword is transmitted, one or more number of transmitted bits will be reversed (0 to 1 or vice versa) due to transmission impairments. Thus errors are introduced.

Error detection techniques :

- Some of the most important error detection methods are as shown in Fig. 8.3.1.

**(G-192) Fig. 8.3.1 : Classification of error detection techniques**

- When a code word is transmitted, one or more number of transmitted bits will be reversed (0 to 1 or vice versa) due to transmission impairments.
- Thus errors will be introduced.
- It is possible for the receiver to detect these errors if the received codeword (corrupted) is not one of the valid code words.
- When the errors are introduced, the distance between the transmitted and received code words will be equal to the number of errors as illustrated in Fig. 8.3.1(a).

Transmitted codeword :	<table><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>	1	0	1	0	1	1	0	0	<table><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>	1	1	1	0	1	0	1	0
1	0	1	0	1	1	0	0											
1	1	1	0	1	0	1	0											
Received codeword :	<table><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>	1	1	1	0	1	1	0	0	<table><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr></table>	0	1	1	1	1	0	1	1
1	1	1	0	1	1	0	0											
0	1	1	1	1	0	1	1											
Number of Errors :	1	2																
Distance :	1	2																

(E-1730) Fig. 8.3.1(a)

- Hence to detect the errors at the receiver, the valid code words should be separated by a distance of more than 1.
- Otherwise the incorrect received code words will also be treated as some other valid code words and the error detection will be impossible.
- The number of errors that can be detected depends on the distance between any two valid code words.

8.3.1 Parity Checking : SPPU : Feb. 16, Dec. 18**University Questions**

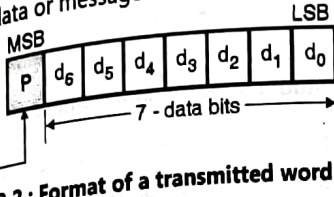
- Q. 1** Write a short note on single parity check codes. (Feb. 16, 4 Marks)
- Q. 2** State channel coding theorem. What are parity bits ? (Dec. 18, 6 Marks)

Definition of parity bit :

- A parity bit or a check bit is a bit added to a string of binary bits to ensure that the total number of 1-bit in the string including the parity bit is either even or odd.

Addition of parity bit :

- The simplest technique for detecting errors is to add an extra bit known as parity bit to each word being transmitted.
- As shown in Fig. 8.3.2, generally the MSB of an 8-bit word is used as the parity bit and the remaining 7 bits are used as data or message bits.



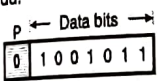
(L-309) Fig. 8.3.2 : Format of a transmitted word with parity bit

Type of parity :

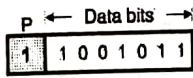
- The parity of the 8-bit transmitted word can be either even parity or odd parity.
- Even parity means the number of 1's in the given word including the parity bit should be even (2, 4, 6...).
- Odd parity means the number of 1's in the given word including the parity bit should be odd (1, 3, 5...).

Use of Parity Bit to Decide Parity :

- The parity bit can be set to 0 or 1 depending on the type of parity required.
- For odd parity this bit is set to 1 or 0 at the transmitter such that the number of "1 bits" in the entire word is odd.



(a) Inclusion of a parity bit to obtain an even parity

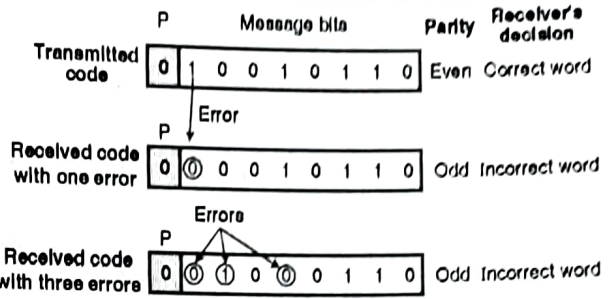


(b) Inclusion of a parity bit to obtain the odd parity

(L-310) Fig. 8.3.3

How does error detection take place ?

- The parity checking at the receiver can detect the presence of an error if the parity of the received signal is different from the expected parity.
- That means if it is known that the parity of the transmitted signal is always going to be "even" and if the received signal has an odd parity then the receiver can conclude that the received signal is not correct.
- This is as shown in Fig. 8.3.4.
- If a single error or an odd number of bits change due to errors introduced during transmission the parity of the code word will change.

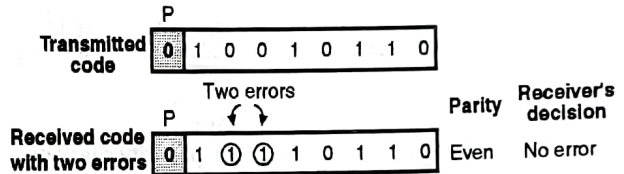


(L-311) Fig. 8.3.4 : The receiver detects the presence of error if the number of errors is odd i.e. 1, 3, 5

- If presence of error is detected then the receiver will ignore the received byte and request for the retransmission of the same byte to the transmitter.

When does parity checking fail to detect errors ?

- If the number of errors introduced in the transmitted code is two or any even number, then the parity of the received code word will not change.
- It will still remain even as shown in Fig. 8.3.5 and the receiver will fail to detect the presence of errors.



(L-312) Fig. 8.3.5 : The receiver cannot detect the presence of error if the number of errors is even i.e. 2, 4, 6

Conclusions :

- Double or any even number of errors in the received word will not change the parity. Therefore even number of errors will be unnoticed.
- If one or odd number of errors occur then the parity of the received word will be different from the parity of transmitted signal. Thus error is noticed. However this error can neither be located nor be corrected.

Limitations of parity checking :

- Following are the limitations of parity check technique :
 - It is not suitable for detection of multiple errors (two, four, six etc).
 - The other limitation of parity checking method is that it cannot reveal the location of erroneous bit.
 - It cannot correct the error either.

8.3.2 Two Dimensional Parity Check (Block Parity) :

SPPU : Dec. 05, May 06, May 16

University Questions

- Q. 1 Explain the following error detection method with suitable examples. Comment on the performance of vertical redundancy check method.

(Dec. 05, May 06, 10 Marks)



Q. 2 Explain two dimensional parity check.

(May 16, 7 Marks)

Block of Data :

- When a large number of binary words are being transmitted or received in succession, the resulting collection of bits is considered as a **block of data**, with rows and columns as shown in Fig. 8.3.6.

Characters	C	O	M	P	U	T	E	R	
7 bit ASCII codes (Message bits)	b ₁	1	1	1	0	1	0	1	0
	b ₂	1	1	0	0	0	0	0	1
	b ₃	0	1	1	0	1	1	1	0
	b ₄	0	1	1	0	0	0	0	0
VRC bits (even parity)	b ₅	0	0	0	1	1	1	0	1
	b ₆	0	0	0	0	0	0	0	0
	b ₇	1	1	1	1	1	1	1	1
→		1	1	0	0	0	1	1	1

These bits will make the parity of each column even

These bits will make the parity of each row even

← LRC bits (even parity)

(L-315) Fig. 8.3.6 : Vertical and longitudinal parity check bits

LRC and VRC Bits :

- The parity bits are produced for each row and column of such block of data.
- The two sets of parity bits so generated are known as :
 - Longitudinal Redundancy Check (LRC) bits
 - Vertical Redundancy Check (VRC) bits.
- The LRC bits indicate the parity of rows and VRC bits indicate the parity of columns as shown in Fig. 8.3.6.

The Vertical Redundancy Check (VRC) Bits :

- As shown in Fig. 8.3.6 the VRC bits are parity bits associated with the ASCII code of each character.
- Each VRC bit will make the parity of its corresponding column "an even parity".
- For example consider column 1 corresponding to character "C".
- The ASCII code for the character C is as follows:

$$C = b_7 \dots b_1 = 1000011$$
- Thus it has three 1's. Therefore, the 8th bit which is a VRC bit is made "1" in order to make the parity even.
- Similarly the other VRC bits are found as shown in Fig. 8.3.6.

Character	C
b ₁	1
b ₂	1
b ₃	0
b ₄	0
b ₅	0
b ₆	0
b ₇	1
VRC bit →	1

← Column - 1 of the data block

← VRC bit = 1 to make the parity of first column even

(G-1944)

The Longitudinal Redundancy Check (LRC) Bits :

- The LRC bits are parity bits associated with the rows of the data block of Fig. 8.3.6.
- Each LRC bit will make the parity of the corresponding row, an even parity.
- For example, consider row 1 of Fig. 8.3.6 as follows.

Row 1 :

b ₁	1	1	1	0	1	0	1	0	1
----------------	---	---	---	---	---	---	---	---	---

 ← LRC bit to make parity even

(G-1945)

- The contents of first row are (1 1 1 0 1 0 1 0), which has five 1's.
- Hence, in order to make the parity of the first row even, we select the LRC bit = 1.

How to locate the error ?

- Even a single error in any bit will result in an incorrect "LRC" in one of the rows and an incorrect VRC in one of the columns.
- The bit which is common to the row and column is the bit in error.
- However there is still a limitation on the block parity code, which is that, multiple errors in rows and columns can be only detected but they cannot be corrected.
- This is because, it is not possible to locate the bits which are in error.
- This will be clear when you will solve the following example.

Ex. 8.3.1 : The following bit stream is encoded using VRC, LRC and even parity. Locate and correct the error if it is present.

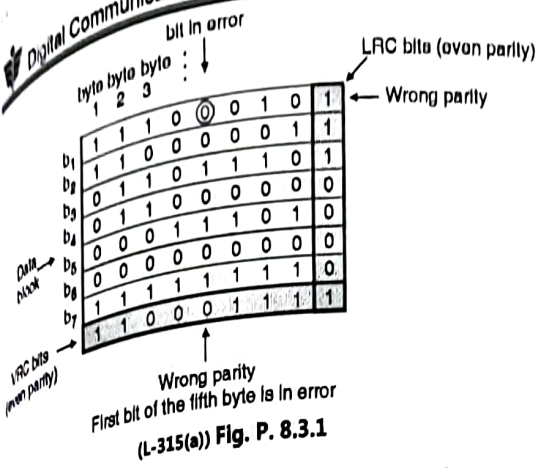
```

11000011 11110011
10110010 00001010
00101010 00101011
10100011 01001011
11100001

```

Soln. :

- Fig. P. 8.3.1 shows the received data block along with the LRC and VRC bits.



- Note the parity bits corresponding to row 1 and column 5 indicate wrong parity. Therefore the fifth bit in the first row (encircled bit) is incorrect.
- Thus using VRC and LRC, it is possible to locate and correct the bits in error.

8.3.3 Checksum Error Detection :

Definition :

- A checksum is a small-sized datum derived from a block of digital data for the purpose of detecting errors that may have been introduced during its transmission or storage.
- Checksum is the last error detection method. It is used in the Internet by many protocols.
- The concept of checksum is based on the principle of redundancy.
- This is very similar to linear block codes and cyclic codes.
- Some protocols still use checksum for error detection but in the recent days CRC is more preferred.
- Hence CRC is fast replacing the checksum.
- Checksum is used at the data link layer level and CRC is used in some layers other than the data link layer.
- Simple parity cannot detect two or even number of errors within the same word.
- One way to overcome this problem is to use a sort of two dimensional parity.

Calculation of checksum :

- As each word is transmitted, it is added to the previously sent word and the sum is retained at the transmitter as shown in Fig. 8.3.7.

$$\begin{array}{r}
 \text{Word A : } 10110111 \\
 + \\
 \text{Word B : } 00100010 \\
 \hline
 \text{Sum : } 11011001
 \end{array}$$

(L-898) Fig. 8.3.7 : Concept of checksum

- Each successive word is added in this manner to the previous sum.
- At the end of the transmission the sum (called a checksum) up to that time is sent.
- The errors normally occur in burst. The parity check method is not useful in detecting the errors under such conditions.
- The checksum error detection method can be used successfully in detecting such errors.
- In this method a "checksum" is transmitted along with every block of data bytes.
- In this method an eight bit accumulator is used to add 8 bit bytes of a block of data to find the "checksum byte".
- The carries of the MSB are ignored while finding out the checksum byte.
- The generation of checksum will be clear if you refer to the following example.

Ex. 8.3.2 : Find the checksum of the following message.

10110001, 10101011,
00110101, 10100001

Soln. :

Carries	10 1 0 1 1 1 0
	+ 1 0 1 1 0 0 0 1
	+ 1 0 1 0 1 0 1 1
	+ 0 0 1 1 0 1 0 1
	+ 1 0 1 0 0 0 0 1
Checksum byte	0 0 1 1 0 0 1 0

(G-1943)

- Note that the carries of MSB have been ignored while writing the checksum byte.

How to detect error using the checksum byte ?

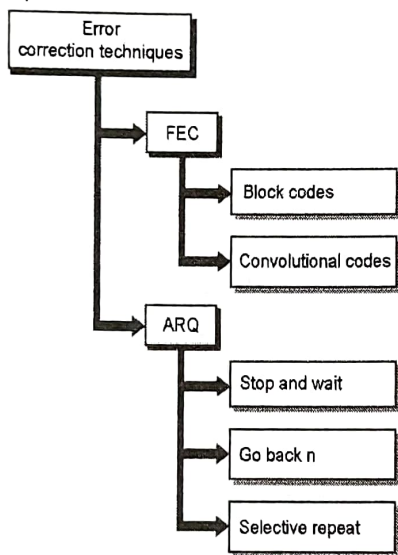
- After transmitting a block of data bytes (say 8-data bytes) the "checksum" byte is also transmitted. The checksum byte is regenerated at the receiver separately by adding the received bytes.
- The regenerated checksum byte is then compared with the transmitted one. If both are identical then there is no error. If they are different then the errors are present in the block of received data bytes.
- Sometimes the 2's complement of the checksum is transmitted instead of the checksum itself. The receiver will accumulate all the bytes including the 2's complement of the checksum. If there is no error, the contents of the accumulator should be zero after accumulation of the 2's complement of the checksum byte.

Advantage of the Checksum Method :

- The advantage of this method over the simple parity checking method is that the data bits are "mixed up" due to the 8 bit addition.
- Therefore checksum represents the overall data block.
- In checksum therefore, there is 255 to 1 chance of detecting random errors.

8.4 Error Correction Techniques :

- The two most important techniques used for error correction are as follows :
 1. Forward Error Correction (FEC)
 2. Automatic request for retransmission (ARQ).
- Fig. 8.4.1 shows the classification of error correction techniques.



(E-2102)Fig. 8.4.1 : Classification of error correction techniques

1. FEC techniques :

- FEC techniques employ special codes that allow the receiver to detect and correct a limited number of errors without referring to the transmitter.
- Some of the codes employed by the FEC techniques are as follows :
 1. Linear block codes.
 2. Convolutional coding.
 3. Hamming codes.
 4. Cyclic codes.

2. ARQ Techniques :

- In the ARQ system, the receiver can request for the retransmission of the complete or a part of message if it finds some error in the received message.
- This needs an additional channel called feedback channel to send the receiver's request for retransmission.

- In the ARQ system of error control, when an error is detected, a request is made for the retransmission of that signal.
- Therefore a feedback channel is required for sending the request for retransmission.
- The ARQ systems differ from the FEC systems in three important respects. They are as follows :
 1. In ARQ system less number of check bits (parity bits) are required to be sent. This will increase the (k/n) ratio for an (n,k) block code if transmitted using the ARQ system.
 2. A return transmission path and additional hardware in order to implement repeat transmission of code words will be needed.
 3. The bit rate of forward transmission must make allowance for the backward repeat transmission.

8.5 Error-correcting Codes :

- We can classify the error correcting codes on the basis of different parameters.
- One way of classifying codes is :
 1. Linear codes
 2. Non-linear codes.

8.5.1 Linear Codes :**Definition :**

- The linear codes are the codes which have an important property that any two valid code words can be added in modulo-2 adder to produce a third valid code word in the code.

Properties of a linear code :

- A linear code should have the following properties :
 1. The sum of any two valid code words belonging to the code produces another valid code word belonging to the code.
 2. The code word with all elements equal to zero is called as all zero word and it is always a valid code word.
 3. The minimum Hamming distance between two code-words of a linear code is equal to the minimum weight of any non-zero code word.
 4. The difference between any two valid code words will produce another valid code word.
- Note that even though the all zero code word is valid for a code, it does not guarantee that the code is a linear code.
- The presence of an all zero code word is thus a necessary but not a sufficient condition for linearity.

8.5.2 Non-Linear Codes :

Definition :
Non-linear codes are the codes which do not exhibit such a property.
All the practically used codes are linear codes.

Ex. 8.5.1 : Explain linearity property of linear block code with examples.

April 18, 4 Marks

Soln. :
Linearity Property :
A block code is said to be linear if the sum of any two code words produces a valid code word.

Example :
Check if the following code is linear or not.

Code $X = \{0000, 0101, 1010, 1111\}$

Add any two code words, as follows :

$$\begin{array}{r} 0000 \\ + 0101 \\ \hline 0101 \end{array} \quad \begin{array}{r} 0101 \\ + 1010 \\ \hline 1111 \end{array}$$

Valid codeword Valid codeword

$$\begin{array}{r} 0101 \\ + 1111 \\ \hline 1010 \end{array} \quad \begin{array}{r} 1010 \\ + 1111 \\ \hline 0101 \end{array}$$

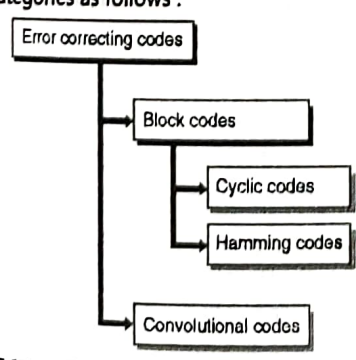
Valid codeword Valid codeword

(G-2475) Fig. P 8.5.1 : Addition of code words

Fig. P. 8.5.1 shows that the given code is a linear block code.

8.5.3 Classification Based on the Functioning :

The error correcting codes are broadly classified into two categories as follows :



(E-173(a)) Fig. 8.5.1 : Classification of codes

The major difference between the two is that the block codes don't need memory whereas convolutional codes do need memory.

Block codes :

- Block codes can be of two types Linear block codes and nonlinear block codes
- Almost all the block codes used today are linear block codes.
- The non-linear block codes are not used widely as their analysis and implementation is difficult.

Definition :

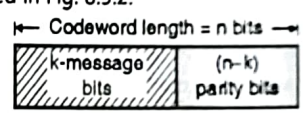
- A linear block code is defined as that code in which the exclusive OR (Modulo - 2 addition) of two valid code words would produce an another valid code word

Examples of linear block codes :

- Some of the examples of linear block codes are as follows:
1. Cyclic codes 2. Hamming codes

Code word structure :

- The code word structure of an 'n' bit linear block code is illustrated in Fig. 8.5.2.

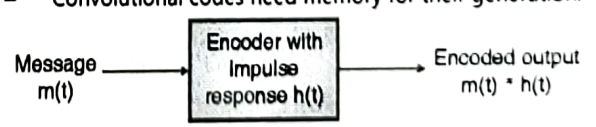


(E-174)Fig. 8.5.2 : Structure of n bit block code word

- The n bit code word consist of "k" message bits followed by (n - k) parity bits.
- The block codes do not need memory.

Convolutional codes :

- In convolutional codes, the code words are generated by discrete time convolution of the input sequence with the impulse response of the encoder.
- Convolutional codes need memory for their generation.



(E-175)Fig. 8.5.3 : Convolutional codes

- In the convolutional code unlike a block code, the channel encoder accepts message bits as a continuous sequence and generates a continuous sequence of encoded bits at its output.
- The major difference between the block codes and convolutional codes is that the convolutional codes need memory whereas the block codes don't.
- Both the types of codes are linear codes.

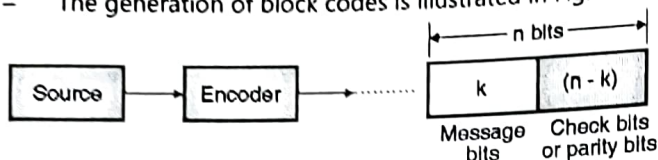
8.6 Linear Block Codes :

Definition :

- In coding theory, a linear code is an error correcting code for which any linear combination of code words is also a code word.

8.6.1 Generation of block code (LBC) :

- The generation of block codes is illustrated in Fig. 8.6.1.

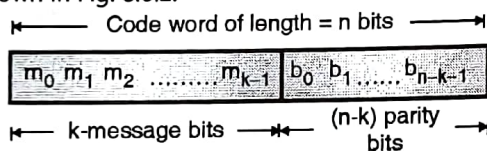


(G-205) Fig. 8.6.1 : Generation of an n bit linear block code

- A channel encoder accepts the information in the form of blocks of k successive bits and adds (n - k) parity bits or check bits at the end of each such block (of k message bits), to generate an (n, k) block code, as shown in Fig. 8.6.1.
- As these bits do not contain any information, they are called as "redundant" bits.
- These are also called as check bits or parity bits.
- It is important to note that the (n - k) parity bits are related algebraically to the "k" message bits.

8.6.2 Code Word Structure of LBC:

- The code word structure of a linear block code is as shown in Fig. 8.6.2.



(E-187) Fig. 8.6.2 : Structure of the code word for a linear block code

- Important points about the code word structure are as follows :
- A code word consists of "k" message bits which are denoted by $m_0, m_1, \dots, m_{k-1}$, and (n - k) : parity bits denoted by $b_0, b_1, \dots, b_{n-k-1}$.
- The sequence of message bits is applied to a linear block encoder to produce an "n" bit code word.
- The elements of this code word are denoted as $x_0, x_1, \dots, x_{n-1}$.
- As shown in Fig. 8.6.2, the first k bits of the code word are identical to the corresponding message bits ($m_0, m_1, \dots$) and the next (n - k) bits are identical to the corresponding parity bits ($b_0, b_1, \dots$).
- We can express this mathematically as :

$$x_i = \begin{cases} m_i & i = 0, 1, \dots, k-1 \\ b_{i-k} & i = k, k+1, \dots, n-1 \end{cases} \quad \dots(8.6.1)$$
- The (n - k) parity bits are "linear sums" of the k message bits as will be discussed later on.

- The code word represented by Equation (8.6.1) can be mathematically represented as :

$$X = [M : B]$$

where M = k - message bits

and B = (n - k), parity bits

...(8.6.2)

8.6.3 Matrix Representation of LBC :

- In this technique, matrices are used for obtaining the code words.
- The generation of code words in the matrix form is as follows :

$$[X]_{1 \times n} = [M]_{1 \times k} [G]_{k \times n}$$

...(8.6.3)

where $[X]_{1 \times n}$ = Code word matrix of 1 x n size.

$[M]_{1 \times k}$ = Message word matrix of 1 x k size.

and $[G]_{k \times n}$ = Generator matrix of k x n size.

8.6.4 Generator Matrix of LBC :

- The generator matrix will be dependent on the type of linear block code used.
- The generator matrix is generally represented as :

$$[G] = [I_k | P]_{k \times n}$$

...(8.6.4)

where I_k = k x k identity matrix.

and P = k x (n - k) coefficient matrix.

- The k x k identity matrix and k x (n - k) coefficient matrix is as given below :

$$\therefore I_k = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}_{k \times k} \quad \text{and } P = \begin{bmatrix} P_{00} & P_{01} & \dots & P_{0,k-1} \\ P_{10} & P_{11} & \dots & P_{1,k-1} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n-k,0} & P_{n-k,1} & \dots & P_{n-k,k-1} \end{bmatrix}_{k \times (n-k)}$$

...(8.6.5)

- The rows of generator matrix G are linearly independent.
- The linear combination of the rows can be used to generate the code words.
- Generator matrix is a (k x n) matrix. For a given linear block code we can have different generator matrices.
- In other words there is no unique generator matrix for a given linear block code.
- A generator matrix converts (encodes) a message vector of length k to a code vector of length n.
- As stated earlier a code word can be obtained from the generator matrix as follows :

$$X = MG$$
- The generator matrix represents the linear block codes very efficiently.
- The n x k matrix can generate q^k code words.

Hence instead of having a large look up table of q^k code words we can simply have a generator matrix. This generator matrix saves a large space in case of large codes.

How to obtain the parity (check) bits ?

The parity bits corresponding to each message word can be obtained as stated below :

$$B = M \times P \quad \dots(8.6.6)$$

Where P represents the coefficient matrix.

The parity vector can be obtained as, $B = MP \quad \dots(8.6.7)$

Substituting the matrix forms of B, M and P into Equation (8.6.7) we get,

$$\begin{bmatrix} b_0 & b_1 & \dots & b_{n-k} \end{bmatrix}_{1 \times n-k} = \begin{bmatrix} m_0 & m_1 & \dots & m_k \end{bmatrix}_{1 \times k} \times \begin{bmatrix} P_{00} & P_{10} & \dots & P_{n-k,0} \\ P_{01} & P_{11} & \dots & P_{n-k,1} \\ \vdots & \vdots & \ddots & \vdots \\ P_{0,k-1} & P_{1,k-1} & \dots & P_{n-k,k-1} \end{bmatrix}_{k \times (n-k)}$$

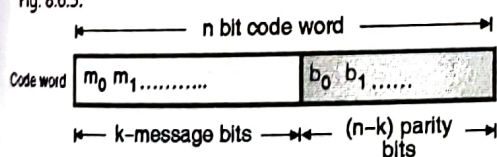
$$(E-1798) \quad \dots(8.6.8)$$

By solving the above matrix equation we can get the parity vectors as follows :

$$\begin{aligned} b_0 &= m_0 P_{00} \oplus m_1 P_{01} \oplus m_2 P_{02} \oplus \dots \oplus m_k P_{0,k-1} \\ b_1 &= m_0 P_{10} \oplus m_1 P_{11} \oplus m_2 P_{12} \oplus \dots \oplus m_k P_{1,k-1} \\ b_2 &= m_0 P_{20} \oplus m_1 P_{21} \oplus m_2 P_{22} \oplus \dots \oplus m_k P_{2,k-1} \end{aligned} \quad \dots(8.6.9)$$

Similarly we can obtain the expressions for the remaining parity bits.

These parity bits (check bits) are appended to the message bits to obtain the code word as shown in Fig. 8.6.3.



(E-1050) Fig. 8.6.3 : Parity bits are appended to the message bits to obtain a code word of linear block code

8.6.5 Generation of Code Words Using Generator Matrix :

If the generator matrix is given along with a message word, then, we can follow the procedure given below to obtain the code words.

Procedure to obtain code words :

Step 1 : From the generator matrix, separate out the coefficient matrix P.

Step 2 : Obtain the parity bits for each message word as $[B] = [M] [P]$.

Step 3 : Append these parity bits after the message bits to get the corresponding code words.

- This procedure will be clear as you solve the following example.

Ex. 8.6.1 : The generator matrix for a (6, 3) block code is given below. Find all the code words of this code.

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Soln. :

- The general pattern of the block codes is (n, k) hence in this case $n = 6$ and $k = 3$.

- That means the message block size k is 3 and the length of code word i.e. n is 6.

Step 1 : Separate the identity matrix I and co-efficient matrix P :

- Referring to Equation (8.6.4), the generator matrix is given by :

$$G = [I_k | P]$$

- Comparing this with the given generator matrix we get,

$$I_k = I_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{and } P = P_{3 \times 3} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

- As the size of message block is $k = 3$, there are eight possible message blocks : (0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1).

Step 2 : Relation between parity vectors B, message vectors M and the coefficient matrix P is as follows :

$$[b_0, b_1, b_2] = [m_0, m_1, m_2] \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad \dots(1)$$

Step 3 : Obtain parity words for all the message words :

- The solution of Equation (1) is given by,

$$b_0 = (m_0 \times 0) \oplus (m_1 \times 1) \oplus (m_2 \times 1) = m_1 \oplus m_2 \quad \dots(2)$$

$$b_1 = (m_0 \times 1) \oplus (m_1 \times 0) \oplus (m_2 \times 1) = m_0 \oplus m_2 \quad \dots(3)$$

$$b_2 = (m_0 \times 1) \oplus (m_1 \times 1) \oplus (m_2 \times 0) = m_0 \oplus m_1 \quad \dots(4)$$

- By substituting the values of m_0 , m_1 , and m_2 into these equations it is possible for us to obtain the parity bits b_0 , b_1 and b_2 .



- For the message word (m_0, m_1, m_2) = 0 0 0

$$b_0 = m_1 \oplus m_2 = 0 \oplus 0 = 0$$

$$b_1 = m_0 \oplus m_2 = 0 \oplus 0 = 0$$

$$b_2 = m_0 \oplus m_1 = 0 \oplus 0 = 0$$

$$\therefore b_0 b_1 b_2 = (0, 0, 0)$$

- The complete code word for this message word is given by,

$$\text{Codeword} = \begin{array}{|c|c|c|c|c|c|} \hline m_0 & m_1 & m_2 & b_0 & b_1 & b_2 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

← Message →
← Parity →
(E-192)

- For the second message vector i.e. (m_0, m_1, m_2) = 0 0 1

$$b_0 = 0 \oplus 1 = 1$$

$$b_1 = 0 \oplus 1 = 1$$

$$b_2 = 0 \oplus 0 = 0$$

$$\therefore (b_0, b_1, b_2) = (1, 1, 0)$$

- The complete code word for this message word is given by,

$$\text{Codeword} = \begin{array}{|c|c|c|c|c|c|} \hline m_0 & m_1 & m_2 & b_0 & b_1 & b_2 \\ \hline 0 & 0 & 1 & 1 & 1 & 0 \\ \hline \end{array}$$

(E-1988)

- Similarly we can obtain the code words for the remaining message words. All these code words have been given in Table P. 8.6.1.

Table P. 8.6.1 : Code words for (6, 3) block code

Sr. No.	Message words			Parity bits			Complete code word					
	m_0	m_1	m_2	b_0	b_1	b_2	m_0	m_1	m_2	b_0	b_1	b_2
1.	0	0	0	0	0	0	0	0	0	0	0	0
2.	0	0	1	1	1	0	0	0	1	1	1	0
3.	0	1	0	1	0	1	0	1	0	1	0	1
4.	0	1	1	0	1	1	0	1	1	0	1	1
5.	1	0	0	0	1	1	1	0	0	0	1	1
6.	1	0	1	1	0	1	1	0	1	1	0	1
7.	1	1	0	1	1	0	1	1	0	1	1	0
8.	1	1	1	0	0	0	1	1	1	0	0	0

8.6.6 Systematic Form of Generator Matrix :

- A generator matrix written in the form

$$G = [I|P]$$

is called as systematic form or standard form of the generator matrix.

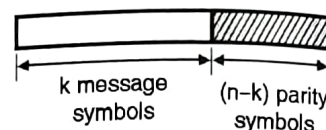
- In the expression, I is a $k \times k$ identity matrix and P is a $k \times (n - k)$ coefficient matrix.

8.6.7 Systematic Code Word :

- An $(n - k)$ systematic code is the one in which the first k symbols of the code word of block length n are the message symbols ($m_0, m_1, \dots$) and the remaining $(n - k)$ symbols are the parity symbols ($b_0, b_1, \dots$) as shown in

Fig. 8.6.4.

Code word of n symbols



(E-1051) Fig. 8.6.4 : Code word structure of a systematic linear code

Ex. 8.6.2 : Parity matrix of (7, 4) LBC is as follows :

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Find the code words for the message :

- i) 0 1 0 1 ii) 1 0 1 0

March 19, 6 Marks

Soln. :

Given : Linear block code, $n = 7$, $k = 4$, $n - k = 3$

Step 1 : Find the code word for message 0101 :

- We know that, parity bits $B = M \times P$

$$\therefore [b_0, b_1, b_2] = [m_0, m_1, m_2, m_3] \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\therefore \text{For } m_0 m_1 m_2 m_3 = 0101$$

$$[b_0, b_1, b_2] = [0101] \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$b_0 = 0 \oplus 1 \oplus 0 \oplus 0 = 1$$

$$b_1 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$b_2 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$\therefore [b_0, b_1, b_2] = [100]$$

$$\therefore \text{Code word for 0101} = 0101100$$

...Ans.

Step 2 : Find the code word for message 1010 :

$$[b_0, b_1, b_2] = [1010] \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\therefore b_0 = 1 \oplus 0 \oplus 1 \oplus 0 = 0$$

$$b_1 = 0 \oplus 0 \oplus 1 \oplus 0 = 1$$

$$b_2 = 1 \oplus 0 \oplus 0 \oplus 0 = 1$$

$$\therefore [b_0, b_1, b_2] = [011]$$

$$\therefore \text{Code word for 1010} = 1010011$$

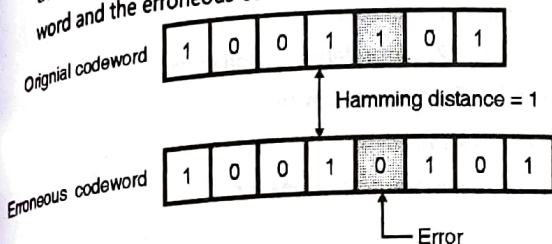
...Ans.

8.7 Decoding of a Linear Block Code :

The channel coding process is used to detect and correct errors when messages are being transmitted over a noisy channel.

The channel noise can randomly alter some of the symbols in the transmitted code word. (from 1 to 0 or from 0 to 1), while travelling from transmitter to receiver.

If only one symbol is changed as shown in Fig. 8.7.1, then the Hamming distance between the original code word and the erroneous code word is 1.



(E-200) Fig. 8.7.1 : Relation between number of errors and Hamming distance

Similarly the noise transforms t symbols (i.e. if t symbols in the received code word are in error), then the Hamming distance of the received word will be " t " with respect to the correct code word.

8.7.1 Error Detection Capability :

An error will be detected as long as it does not transform one code word into another valid code word.

If the minimum distance between the code word is d_{min} then the weight of the error pattern must be d_{min} or more to cause transformation of one code word to another.

The error detection is always possible when the number of transmission errors in a code word is less than the minimum distance d_{min} , because then the erroneous word is not a valid code word.

But when the number of errors is equal to or greater than d_{min} the erroneous code word may correspond to some other valid code word and errors cannot be detected.

The error detection and correction capabilities of a coding technique depends on the minimum distance d_{min} .

A code having a minimum distance d_{min} will detect all the non-zero error patterns of weight less than or equal to $(d_{min} - 1)$.

Hence we can say that upto " S " errors per word can be detected such that

$$d_{min} \geq (S + 1)$$

$$\text{or } S \leq (d_{min} - 1) \quad \dots(8.7.1)$$

- Also note that there is at least one error pattern of weight d_{min} which cannot be detected.
- This happens for the two code words which are the closest.
- It is possible that some error patterns of weight d_{min} or more are detected but all error patterns of weight d_{min} will never be detected.

8.7.2 Error Correcting Capability :

- Let us now see, how many errors can be successfully corrected by a code.
- This will be based on the best possible guess about the original transmitted code word.
- It is logical to guess that a valid code word which is nearest (in terms of Hamming distance) to the received code word in terms of hamming distance must have been actually transmitted.
- This technique is called as **Nearest Neighbour Decoding**.
- But the possible problem with this technique is that it is possible that multiple valid code words could be at the same Hamming distance from the received word.

In this situation the receiver does the following :

- It can randomly pick up one of the equally distant code words on a random basis or
- It can request the transmitter to retransmit the code word again.

To ensure that the received word having at the most t errors, is closest to the original code word, we put the following condition on the minimum distance.

$$d_{min} \geq 2t + 1 \quad \dots(8.7.2)$$

Hence the number errors that can be successfully corrected by a code is given by

$$t \leq \frac{d_{min} - 1}{2} \quad \dots(8.7.3)$$

The error detecting and correcting capability of a code is summarized in Table 8.7.1.

Table 8.7.1 : Role of d_{min} for detection and correction of errors

Detect upto " S " errors per word.	$d_{min} \geq (S + 1)$
Correct upto " t " errors per word.	$d_{min} \geq (2t + 1)$
Correct upto " t " errors and Detect $S > t$ errors per word.	$d_{min} \geq (t + S + 1)$



Ex. 8.7.1 : For a (4, 2) linear block code, the generator matrix is given as :

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Find all code words that can be generated. Comment on error correction capability of the code. **Feb. 16, 6 Marks**

Soln. :

Given : $n = 4, k = 2$

$$G = [I | P]$$

$$\therefore P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\text{Parity bits } [b_0, b_1] = [m_0, m_1] [P]$$

Code word for $m_0, m_1 = 00$:

$$[b_0, b_1] = [0, 0] \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = [0, 0]$$

$$\therefore \text{Codeword} = \underbrace{00}_{\text{Message (E-1706)}} \underbrace{00}_{\text{Parity}}$$

...Ans.

- Similarly we can find the code words for the other messages. Table P. 8.7.1 enlists all the code words.

Table P. 8.7.1 : Code words

Message	Codeword	Hamming Distance
00	0000	0
01	0101	2
10	1011	3
11	1110	3

$\leftarrow d_{\min}$

Step 1 : Error detecting and correcting capability :

- The minimum Hamming distance from Table P. 8.7.1 is,

$$d_{\min} = 2$$

- $\therefore$ Error detecting capability is,

$$S \leq (d_{\min} - 1)$$

$$\therefore S \leq 2 - 1$$

$$\therefore S \leq 1$$

...Ans.

- Thus at the most one error can be detected.

- Number of correctable errors,

$$t \leq \frac{d_{\min} - 1}{2}$$

$$\therefore t \leq \frac{2 - 1}{2}$$

$$\therefore t \leq \frac{1}{2}$$

...Ans.

- Thus no error can be corrected.

Ex. 8.7.2 : For a systematic (6, 3) LBC, the parity matrix is given by,

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

- Find all possible code vectors.
- Find error detecting and correction capabilities.

May 18, 7 Marks

Soln. :

Given : A LBC, $n = 6, k = 3, (n - k) = 3$.

To find : 1. All code words

- Error detecting and correcting capability.

Part I : Find all code words :

- We know that,

$$\text{Parity bits } B = M \times P$$

$$\therefore [b_0, b_1, b_2] = [m_0, m_1, m_2] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

- For $m_0, m_1, m_2 = 001$

$$[b_0, b_1, b_2] = [0, 0, 1] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = 110$$

- Similarly we can obtain the parity check bits and code words for all the messages as given in Table P. 8.7.2.

(G-2472) Table P. 8.7.2

Message bits m_0, m_1, m_2	Parity bits b_0, b_1, b_2	Codewords	Hamming distance
0 0 0	0 0 0	0 0 0 0 0 0	0
0 0 1	1 1 0	0 0 1 1 1 0	3
0 1 0	0 1 1	0 1 0 0 1 1	3
0 1 1	1 0 1	0 1 1 1 0 1	4
1 0 0	1 0 1	1 0 0 1 0 1	3
1 0 1	0 1 1	1 0 1 0 1 1	4
1 1 0	1 1 0	1 1 0 1 1 0	4
1 1 1	0 0 0	1 1 1 0 0 0	3

$\leftarrow d_{\min}$

Part II : Error detecting and correcting capability :

- From Table P. 8.7.2, $d_{\min} = 3$

Error detecting capability :

$$S \leq (d_{\min} - 1)$$

$$\therefore S \leq 2$$

- Hence at the most two errors can be detected.

Error correcting capability :
Number of correctable errors

$$t \leq \frac{d_{\min} - 1}{2}$$

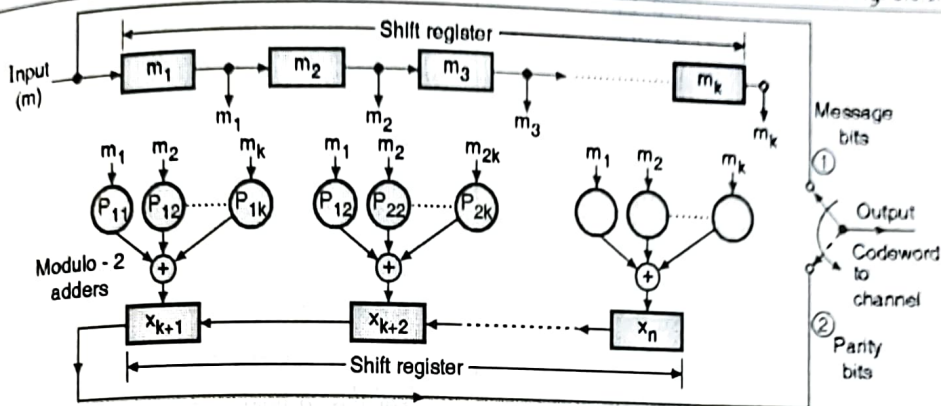
$$t \leq \frac{3 - 1}{2}$$

Hence at the most one error can be corrected

8.8 Encoder Implementation :

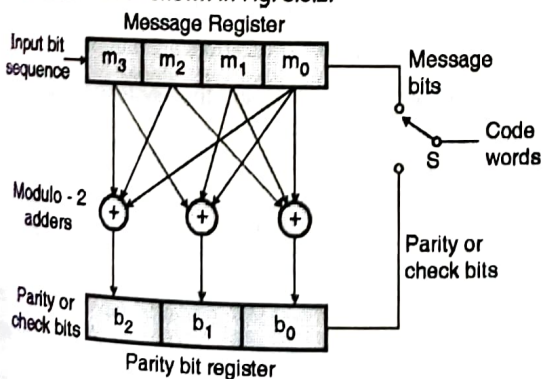
Block diagram :

The general structure of an encoder circuit for (n, k) linear block code is as shown in Fig. 8.8.1



(E-1728) Fig. 8.8.1 : General structure of an encoder for LBC

- The input to this encoder is the k -bit message $m = (m_1, m_2, \dots, m_k)$ and we get the n -bit code word $X = (x_k, x_{k-1}, \dots, x_2, x_1)$ at its output.
- The k message bits that are to be encoded are shifted into the registers one by one.
- These bits are transmitted over the channel simultaneously.
- After shifting all the k message bits into the shift register, the encoder calculates $(n - k)$ parity bits using the modulo-2 arithmetic.
- These parity bits are shifted one by one onto the channel.
- A more precise encoder circuit for a $(7, 4)$ linear block code has been shown in Fig. 8.8.2.



(E-196) Fig. 8.8.2 : Encoder for $(7, 4)$ LBC

- This encoder produces all the code words corresponding to various message words.

- The parity or check bits (b_2, b_1, b_0) are being generated for each message word (m_3, m_2, m_1, m_0) .
- The parity bits are obtained from the message bits by means of modulo-2 adders. (Modulo-2 adders are nothing but EX-OR gates).
- The output switch "S" is first connected to the message register in order to output all the message bits in a code word.
- Then it is connected to the parity bit register to output the corresponding parity bits.
- Thus we get a 7 bit code word at the output of the switch.

Ex. 8.8.1 : For a $(6, 3)$ code, the generator matrix G is given by :

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

- Realize an encoder for this code.
- Verify that this code is a single error-correcting code.

Soln. :

- Obtain the expressions for the parity bits :**

- The parity bits can be obtained by using the expression,

$$B = MP$$

$$\therefore [b_0, b_1, b_2]_{1 \times 3} = [m_0, m_1, m_2]_{1 \times 3} [P]_{3 \times 3}$$

- The parity matrix can be obtained from the generator matrix as follows,



$$G = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array} \right]$$

$$\left| \begin{array}{ccc} \longleftrightarrow & & \longleftrightarrow \\ I_{3 \times 3} & & P_{3 \times 3} \end{array} \right|$$

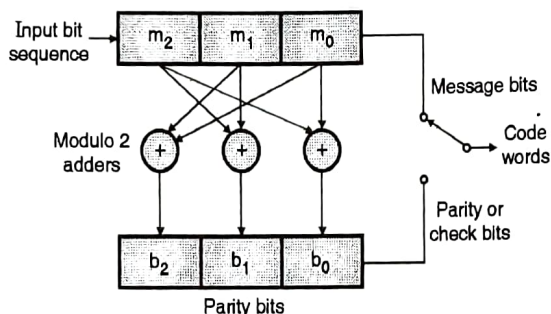
$$\therefore [P] = \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{array} \right]_{3 \times 3}$$

$$\therefore [b_0, b_1, b_2] = [m_0, m_1, m_2] \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{array} \right]$$

$$\left. \begin{aligned} \therefore b_0 &= m_0 \oplus m_2 \\ b_1 &= m_1 \oplus m_2 \\ b_2 &= m_0 \oplus m_1 \end{aligned} \right\} \dots(1)$$

2. Draw the encoder :

- The encoder is obtained to implement the expressions given in Equation (1) as shown in Fig. P. 8.8.1.



(E-208) Fig. P. 8.8.1 : Encoder block diagram

Ex. 8.8.2 : The generator matrix for a (6, 3) block code is given below. Find all code vectors. What is minimum distance of this code ? How many errors this code can detect ?

$$G = \left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array} \right]$$

Dec. 06, 12 Marks

Soln. :

Part 1 :

- For all the code vectors please refer Ex. 8.6.1.

Part 2 : Number of errors detectable :

- The minimum distance of (6, 3) code is $d_{\min} = 3$.

$\therefore$ Number of errors detectable is "s" such that

$$d_{\min} \geq (S + 1)$$

$$\therefore S \leq (d_{\min} - 1)$$

$$\therefore S \leq (3 - 1) \quad \therefore S \leq 2$$

- So at the most 2 errors can be detected.

8.9 Parity Check Matrix of Block Codes :

- One of the aims of a good code design is that encoding and decoding should be fast and efficient.
- Generator matrix ensures the efficient way of generating the linear block codes.
- We simply multiply the message vector with the generator matrix to produce the code words.
- The question is, can we do the same for detecting (decoding) a valid code word ?
- The answer is yes and a matrix used for decoding linear block codes is called as **parity check matrix (H)**, for the given code. A parity check matrix is such that

$$XH^T = 0$$

where X is a valid code word and H^T is transpose of H.

- But $X = MG$

$$\therefore MGH^T = 0$$

- For this to hold true for all the code words, we must have

$$GH^T = 0$$

- The size of the parity check matrix is $(n - k) \times n$. A parity check matrix provides a simple way to find out whether an error is present or not in the received code word.

8.9.1 Error Detection using Parity Check Matrix :

- Multiply the received code word X with the transpose of parity check matrix (H^T).
- If the product is a nonzero vector, it indicates that an error has occurred and if the product is an all zero vector, then it is an indication of no errors in the received code word.
- That means :

$$\therefore XH^T = 0 \quad \dots \text{no errors}$$

$$XH^T \neq 0 \quad \dots \text{errors are present in X}$$

- The generator matrix in the systematic form is expressed as $G = [I|P]$ and the parity check matrix is denoted by,

$$H = [P^T | I] \quad \dots(8.9.1)$$

$$H = [P^T | I_{n-k}] \quad \dots(8.9.2)$$

where P^T = An $(n - k) \times k$ matrix representing the transpose of the coefficient matrix P and I_{n-k} is an $(n - k) \times (n - k)$ identity matrix.

- The transpose of the coefficient matrix is obtained by interchanging the rows and columns of the coefficient matrix P.

$$P^T = \begin{bmatrix} P_{00} & P_{01} & \dots & P_{0,k-1} \\ P_{10} & P_{11} & \dots & P_{1,k-1} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n-k,0} & P_{n-k,1} & \dots & P_{n-k,k-1} \end{bmatrix} \quad (n-k) \times k \quad \dots(8.9.3)$$

(E-526)

Therefore the parity check matrix H is given by,

$$H = \begin{bmatrix} P_{00} & P_{01} & \dots & P_{0,k-1} & 1 & 0 & 0 & \dots & 0 \\ P_{10} & P_{11} & \dots & P_{1,k-1} & 0 & 1 & 0 & \dots & 0 \\ P_{20} & P_{21} & \dots & P_{2,k-1} & 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ P_{n-k,0} & P_{n-k,1} & \dots & P_{n-k,k-1} & 0 & 0 & 0 & \dots & 1 \end{bmatrix} \quad (n-k) \times k \quad \dots(8.9.4)$$

(E-527)

The matrix H is called as the parity check matrix. If generator matrix G has been given then we can obtain the parity check matrix and vice versa.

We have stated earlier that a given code have multiple generator matrices.

Therefore a given code can have multiple parity check matrices as well. In other words, the parity check matrix also **will not be unique, for a given code**.

It is possible to obtain a parity check matrix H if the generator matrix G has been given to us.

8.9.2 Relation between H and $d_{\min}$:

We have seen that

$$X H^T = 0 \quad \dots \text{for no errors in } X$$

For this to happen, vector X must have 1s in such positions that the corresponding rows of H^T add up to yield a zero vector 0.

But the number of 1s in a code word pertains to its Hamming Weight.

Hence the minimum distance $d_{\min}$ of a linear block code is given by the minimum number of rows of H^T (or columns of H) whose sum is equal to the zero vector.

This is the relation between H and $d_{\min}$.

Ex. 8.9.1 : Find the parity check matrix for decoding linear block code if generator matrix is given as :

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

Feb. 16, 4 Marks

Soln. :

Step 1 : Separate the P matrix from G :

The systematic form of generator matrix is given by,

$$G = [I_k | P]$$

The given code is (6, 3) block code.

$$\therefore n = 6, k = 3.$$

So identity matrix will be 3×3 matrix.

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Step 2 : Obtain the parity check matrix :

Parity check matrix is of size $(n - k) \times n$ i.e. here it will be 3×6 matrix.

$$H = [P^T | I]$$

$$\text{But } P^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \text{ and } I_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

...Ans.

Ex. 8.9.2 : Consider a (7, 4) linear block code whose generator matrix is given by :

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

1. Obtain the code word for the message 0101.

2. Find the parity check matrix.

Dec. 08, 8 Marks, Dec. 11, 12 Marks

Soln. :

Step 1 : Obtain the P matrix :

(E-532)

$$\text{Given generator matrix } G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$\underbrace{\hspace{1.5cm}}_{I_{4 \times 4}} \quad \underbrace{\hspace{1.5cm}}_{P_{4 \times 3}}$



- Therefore the P matrix is given by,

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{4 \times 3} \quad \dots(1)$$

Step 2 : To obtain the parity (check) bits :

- For the message [0 1 0 1], the parity bits can be obtained by using the expression.

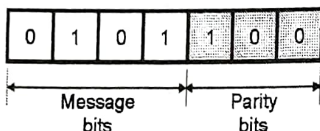
$$B = MP$$

$$\therefore [b_0 \ b_1 \ b_2]_{1 \times 3} = [0 \ 1 \ 0 \ 1] \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{4 \times 3} = [1 \ 0 \ 0]$$

- Therefore the complete code word, for the message word 0 1 0 1 is

(E-533)

Complete code word =



- Similarly we can obtain the code words for the remaining message words.
- All the message vectors, the corresponding parity bits and code words are given in Table P. 8.9.2.
- The code weights are also given in the Table P. 8.9.2.

Table P. 8.9.2 : Code vectors for all the message vectors

Sr. No.	Message vector, M				Parity bits, B			Code words, X						
	m_3	m_2	m_1	m_0	b_2	b_1	b_0	X_6	X_5	X_4	X_3	X_2	X_1	X_0
1.	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2.	0	0	0	1	1	0	1	0	0	0	1	1	0	1
3.	0	0	1	0	0	1	1	0	0	1	0	0	1	1
4.	0	0	1	1	0	1	0	0	0	1	1	0	1	0
5.	0	1	0	0	0	1	1	0	1	0	0	0	1	1
6.	0	1	0	1	1	1	0	0	1	0	1	1	1	0
7.	0	1	1	0	1	0	0	0	1	1	0	1	0	0
8.	0	1	1	1	0	0	1	0	1	1	1	0	0	1
9.	1	0	0	0	1	1	0	1	0	0	0	1	1	0
10.	1	0	0	1	0	1	1	1	0	0	1	0	1	1
11.	1	0	1	0	0	0	1	1	0	1	0	0	0	1

Sr. No.	Message vector, M				Parity bits, B			Code words, X						
	m_3	m_2	m_1	m_0	b_2	b_1	b_0	X_6	X_5	X_4	X_3	X_2	X_1	X_0
12.	1	0	1	1	1	0	0	1	0	1	1	1	0	0
13.	1	1	0	0	1	0	1	1	1	0	0	1	0	1
14.	1	1	0	1	0	0	0	1	1	0	1	0	0	0
15.	1	1	1	0	0	1	0	1	1	1	0	0	1	0
16.	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Step 3 : To obtain the parity check matrix :

- The parity check matrix [H] is a 3×7 matrix,

$$H = [P^T : I_{n-k}] = [P^T : I_3]$$

- The transpose matrix P^T is given by :

$$P^T = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}_{3 \times 4}$$

$$\therefore H = [P^T : I_{3 \times 3}]$$

$$= \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}_{3 \times 7} \quad \dots \text{Ans.}$$

- This is the required parity check matrix.

Ex. 8.9.3 : The parity check matrix of a particular (7, 4) linear block code is given by,

$$[H] = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

- Find the generator matrix (G).
- List all the code vectors.
- What is the minimum distance between the code vectors ?
- How many errors can be detected ? How many can be corrected ?

Dec. 15, 7 Marks

Soln. :

Given : $n = 7, k = 4, (n - k) = 3$

Step 1 : Find the generator matrix G :

$$\text{Given... } H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$\xleftarrow{\quad P^T \quad} \quad \xleftarrow{\quad I_3 \quad} \quad \text{(E-1707)}$

- The generator matrix is a $(k \times n)$ i.e. (4×7) matrix as follows :

$$G = [I_4 : P] \quad (\text{E-1708})$$

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$\underbrace{\quad\quad\quad}_{I_4} \quad \underbrace{\quad\quad\quad}_P$

...Ans.

Step 2 : Find all the code words :

The parity bits for the message [0 0 0 1] is obtained as follows :

$$B = M \times P$$

$$\therefore [b_0 \ b_1 \ b_2] = [0 \ 0 \ 0 \ 1] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$= [0 \ 1 \ 1]$$

...Ans.

$$\therefore \text{Codeword} = 0001011$$

For message [0 0 1 0] the parity bits are,

$$\therefore [b_0 \ b_1 \ b_2] = [0 \ 0 \ 1 \ 0] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$= [1 \ 0 \ 1]$$

...Ans.

$$\therefore \text{Codeword} = 0010101$$

...Ans.

- Similarly we can obtain the remaining code words.
- All the code words are listed in Table P. 8.9.3. The Hamming distance is equal to the number of 1s in a code word.

Table P. 8.9.3

Message	Codeword	Hamming Distance
0000	0000 000	0
0001	0001 011	3
0010	0010 101	3
0011	0011 101	4
0100	0100 101	3
0101	0101 101	4
0110	0110 011	4
0111	0111 000	3
1000	1000 111	4
1001	1001 100	3
1010	1010 010	3
1011	1011 001	4

Message	Codeword	Hamming Distance
1100	1100 001	3
1101	1101 010	4
1110	1110 100	4
1111	1111 111	7

Step 3 : Find $d_{\min}$:

- From Table P. 8.9.3, $d_{\min} = 3$.

Step 4 : Error detection and correction capability :

- Number of detectable errors is,

$$S \leq (d_{\min} - 1)$$

$$\therefore S \leq 2$$

...Ans.

- Thus at the most two errors can be detected.

- Number of correctable errors will be,

$$t \leq \frac{d_{\min} - 1}{2}$$

$$\therefore t \leq 1$$

...Ans.

- Thus at the most one error can be corrected.

8.10 Decoding LBC using Parity Check Matrix :

- In section 8.9 we have defined the parity check matrix H for a code.
- Let us now see, how to use the parity check matrix for decoding of the received code words.
- Let the received code word be denoted by Y .
- The receiver does not know the transmitted code word but it knows the coefficient matrix $[P]$ which was used for the generation of these code words.
- The $[P]$ matrix is used for checking the message bits using the parity bits (check bits) received in the code word.

- The decoding procedure is as follows :

Step 1 : Obtain the parity check matrix H :

- The parity check matrix is given by,

$$[H] = [P^T | I_{(n-k)}] \quad \dots(8.10.1)$$

where P^T = Transpose of coefficient matrix.

and $I_{(n-k)}$ = Identity matrix of the order $(n - k)$.

where $(n - k)$ = Number of parity bits.

Step 2 : Take the transpose of parity check matrix :

$$\therefore H^T = \begin{bmatrix} P \\ \dots \\ I_{n-k} \end{bmatrix}_{n \times (n-k)}$$



- The transpose of parity check matrix H^T exhibits a very important property as follows :

$$YH^T = (0, 0, \dots, 0)$$

If there is no error in the received code word Y .

$$\text{and } YH^T \neq (0, 0, \dots, 0)$$

If error exists in the received code word Y .

- The detection procedure will be clear from the following example :

Ex. 8.10.1 : For a (6,3) linear block code, the coefficient matrix $[P]$ is as follows :

$$[P] = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}_{3 \times 3}$$

- The received code words at the receiver are :

$$1. \ 001110 \quad 2. \ 111011$$

- Check whether they are correct or contain errors.

Soln. :

Given : $n = 6, k = 3, (n - k) = 3$

Step 1 : Obtain the parity check matrix H :

- The parity check matrix is given by

$$H = [P^T | I_{n-k}] = [P^T : I_3] \quad (\text{E-1709})$$

$$H = \left[\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right]$$

$\underbrace{\quad\quad\quad}_{P^T} \quad \underbrace{\quad\quad\quad}_{I_3}$

Step 2 : Obtain the transpose of H i.e. H^T : (E-1710)

$$H^T = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 3 : For each code word obtain YH^T :

- The first received code word $Y_1 = 001110$

$$\therefore Y_1 H^T = [001110] \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = [000] \quad (\text{E-1711})$$

$$\therefore Y_1 H^T = [0, 0, 0]$$

- As the received code word is a valid code word the product $Y_1 H^T$ results into $[0, 0, 0]$ which indicates that there is **no error present** in the received code word.

- The second received code word $Y_2 = [111011]$

- Note that this code word is not present in Table P. 8.6.1.

- So it is a code word containing errors.

$$\therefore Y_2 H^T = [111011] \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = [011] \quad (\text{E-1712})$$

$$\therefore Y_2 H^T = [011]$$

- Since $Y_2 H^T \neq [0, 0, 0]$, it indicates that the received code word consists of some errors.

- That means the received code word is wrong. Thus the presence of errors has been detected using the transpose of parity check matrix.

Ex. 8.10.2 : For a code vector $X = (0 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0)$ and the parity check matrix H given below, prove that, $XH^T = (0, 0, \dots, 0)$ if the transmitted code word is received without any error

$$H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}_{3 \times 7}$$

Soln. :

Procedure :

- Find the transpose matrix H^T .
- Calculate the product XH^T .

Step 1 : Find H^T :

- The transpose of the given parity matrix H , is obtained by interchanging the rows and columns as,

$$H^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{7 \times 3}$$

Step 2 : Calculate the product XH^T :

- The product XH^T is given by,

$$XH^T = [0111000]_{1 \times 7} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{7 \times 3} = [000]$$

$$(\leftarrow H^T \rightarrow)$$

$$\therefore XH^T = [0, 0, 0]_{1 \times 3}$$

Thus we have proved that for a valid code word, the product $XH^T = (0, 0, 0)$.

8.10.1 Shannon's Limit :

It is always desirable to have codes of large block lengths because the code rates of larger codes perform closer to the Shannon Limit.

One of the important aims of coding theory is to develop an efficient decoding strategy.

In order to construct a decoder which will work in real time the decoding scheme must satisfy the following requirements :

1. We should be able to realize the decoding scheme in terms of memory requirement.
 2. The scheme should be realizable in terms of computational load.
- Both these requirements are fulfilled by a technique called **syndrome decoding**.

8.11 Standard Array and Syndrome Decoding :

SPPU : May 16

University Questions

Q.1 What is a standard array decoding ? Explain with suitable example. (May 16, 7 Marks)

- The two important functions of the decoder are :
 1. Error detection in the received codeword.
 2. Error correction.
- The decoding of linear block codes is done with the help of a special technique called **syndrome decoding**.
- The error correcting capability of a code is determined by the value of minimum Hamming distance d_{min} . Let us now discuss how to correct these errors.
- We can use the generator matrix (G) for the generation of a code if the message is given. On the other hand the received message can be decoded using the parity check matrix.
- Let Y represent the received code vector. It may or may not be identical to the transmitted code vector X.

8.11.1 Error Vector (E) :

Let 'E' represent the **error vector** which defines the corresponding error pattern.

The relation between X, Y and E is as follows :

$$Y = X \oplus E \quad \dots(8.11.1)$$

- In the absence of errors (received codeword same as the transmitted one) the error vector 'E' contains all zeros.
- A '1' in the error vector indicates the presence of error in the corresponding location in the received codeword.
- That means,

$$E_i = 1 \quad \dots \text{if an error has occurred in the } i^{\text{th}} \text{ position.}$$

$$= 0 \quad \dots \text{No error}$$

- If the transmitted and received codewords X and Y are n bit long, then the error vector E also will be n-bit long.
- The non-zero elements in the error vector E represent the locations of errors in the received code vector Y as shown in Fig. 8.11.1.

Transmitted codeword X :

0	0	1	1	1	1	0
---	---	---	---	---	---	---

Received codeword Y :

1	0	0	1	0	1	0
---	---	---	---	---	---	---

Errors

Error vector E :

1	0	1	0	1	0	0
---	---	---	---	---	---	---

The nonzero element in E represent the location of errors in received codeword Y

(E-1713) Fig. 8.11.1 : Error vector E along with the transmitted and received code vectors X and Y

- The shaded boxes in Fig. 8.11.1 represent the presence of errors.
- The relation between X, Y and E is as follows :

$$Y = X \oplus E \quad \dots(8.11.2)$$

$$\text{And } X = Y \oplus E \quad \dots(8.11.3)$$

- Thus we can get the correct transmitted codeword X if the erroneous received codeword is added to the error vector (E).

8.11.2 Syndrome Vector (S) :

- We obtain the syndrome vector (S) by multiplying the received code vector (Y) with the H^T matrix.

$$\therefore S = YH^T \quad \dots(8.11.4)$$

- If the received codeword is exactly same as the transmitted one (no error), then the value of syndrome vector is zero.

$$\text{i.e. } S = 0 \quad \dots \text{If no errors present in}$$

the received codeword.

- Received codeword (Y) has an order $(1 \times n)$ and H^T is of the order $n \times n - k$, therefore the syndrome vector (S) will have $(n - k)$ bits.

- If $Y \neq X$ then,

$$S = YH^T = (X \oplus E)H^T = X \cdot H^T \oplus E \cdot H^T$$

$$\therefore S = E \cdot H^T \quad \dots(8.11.5)$$

- This expression shows that the syndrome is dependent on the error pattern E .
- Another important property of syndrome states that :
- All error patterns that differ by a codeword will have the same syndrome.
- This can be proved in the following way.
- Let the number bits per message be equal to " k ".
- Therefore there will be 2^k possible codewords which are denoted by $X_1, X_2, X_3, \dots, X_k$.
- Let E represent an error pattern, which also has 2^k distinct vectors denoted by $E_1, E_2, \dots, E_k$.

$$\therefore E_i = E \oplus X_i \quad \dots(8.11.6)$$

- We will call the set of vectors $\{E_1, E_2, \dots, E_k\}$ as the **coset of the code**.

- For an (n, k) block code, there will be 2^k possible cosets.

- Now,

$$E_i \cdot H^T = (E \oplus X_i) \cdot H^T = E \cdot H^T \oplus X_i \cdot H^T$$

$$= E \cdot H^T = S \quad \dots(8.11.7)$$

- The above expression proves that each coset of the code is characterized by a **unique syndrome**.
- A **coset leader** is defined as the vector that has the minimum weight in the coset.

8.11.3 Standard Array :

SPPU : May 16

University Questions

- Q. 1** What is a standard array decoding ? Explain with suitable example. **(May 16, 7 Marks)**

- We can construct a standard array by using the coset leaders. All the valid code words starting with the all zero code word are written in the first row.
- Then we need to write down all the syndromes in the syndrome column (as shown in Table 8.11.1) and the coset leaders which are actually the error patterns in the coset leader column.
- The remaining column of n-tuples can be written by adding coset leader and the code word in the first row.
- These n-tuples actually correspond to the received code word with the corresponding syndrome written in the syndrome column.

(E-2115) Table 8.11.1 : Standard array for $X = \{000, 111\}$

Syndrome	Coset leader	n- Tuples	
0 0 (no errors)	0 0 0	1 1 1	Code vectors Single errors
1 1	1 0 0	0 1 1	
1 0	0 1 0	1 0 1	
0 1	0 0 1	1 1 0	

8.11.4 Procedure for Syndrome Decoding :

SPPU : March 19

University Questions

- Q. 1** Explain the syndrome decoding operation for (n, k) block code with the help of diagram.

(March 19, 4 Marks)

- The procedure for syndrome decoding is as given below :
 - From the received codeword Y , compute the corresponding syndrome $S = YH^T$.
 - Identify the error pattern (or coset leader) corresponding to syndrome S computed in the first step and call it as E .
 - Obtain the transmitted codeword X as follows :

$$X = Y \oplus E$$
- Following examples illustrate this procedure.

Ex. 8.11.1 : The parity check matrix of a $(7, 3)$ linear code is as follows :

$$H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}_{7 \times 3}$$

Calculate the syndrome vector for single bit errors.

Soln. :

Step 1 : The syndrome vector is given by :

$$S = EH^T = [E]_{1 \times 7} [H^T]_{7 \times 3}$$

$\therefore$ Syndrome vector will be represented by a 1×3 matrix.

$$\therefore [S]_{1 \times 3} = [E]_{1 \times 7} [H^T]_{7 \times 3} \quad \dots(1)$$

Step 2 : Write various error vectors for single bit errors :

- Various error vectors with single bit errors are as shown in Table P. 8.11.1(a).
- The shaded bits represent the locations of errors.

(E-1716) Table P. 8.11.1(a)

Error vectors	Bit in error
0 0 0 0 0 0 0	—
1 0 0 0 0 0 0	First
0 1 0 0 0 0 0	Second
0 0 1 0 0 0 0	Third
0 0 0 1 0 0 0	Fourth
0 0 0 0 1 0 0	Fifth
0 0 0 0 0 1 0	Sixth
0 0 0 0 0 0 1	Seventh

Step 3: Calculate the syndromes corresponding to each error vector:

$$[S]_{1 \times 3} = [E]_{1 \times 7} [H^T]_{7 \times 3}$$

Substituting $[E] = (1000000)$ and H^T we get,

$$[S] = [1000000]_{1 \times 7} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{7 \times 3} = [1 \ 1 \ 1]$$

$\leftarrow H^T \rightarrow$

This is the syndrome for the first bit in error.

2. For the second bit in error,

$$[S] = [0100000] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = [1 \ 1 \ 0]$$

3. Similarly we can obtain the other syndromes as shown in Table P. 8.11.1(b).

(E-1717) Table P. 8.11.1(b) : Syndromes for various error vectors

Sl. No.	Error vectors with single bit errors	Syndrome vector "S"	Relation with H^T
1	0 0 0 0 0 0 0	0 0 0	
2	1 0 0 0 0 0 0	1 1 1	1 st Row of H^T
3	0 1 0 0 0 0 0	1 1 0	2 nd Row of H^T
4	0 0 1 0 0 0 0	1 0 1	3 rd Row of H^T
5	0 0 0 1 0 0 0	0 1 1	4 th Row of H^T
6	0 0 0 0 1 0 0	1 0 0	5 th Row of H^T
7	0 0 0 0 0 1 0	0 1 0	6 th Row of H^T
8	0 0 0 0 0 0 1	0 0 1	7 th Row of H^T

4. Note that the first row of Table P. 8.11.1(b), represent an error vector with no errors. The corresponding syndrome is (0, 0, 0).

5. Table P. 8.11.1(b) also shows that the syndrome vectors are same as the rows of the transpose matrix H^T .

Ex. 8.11.2 : Obtain the code words for the (8, 3) LBC which has the generator matrix of $G = [110100 : 011010 : 101001]$. If code word $C = 101110$ is transmitted and received code word is $r = 001110$. Obtain the correct code word by use of syndrome polynomial.

May 13, 8 Marks

Soln. :

Step 1 : Write the generator matrix :

Given : $G = [110100 : 011010 : 101001]$

It can be written in the matrix form as follows :

$$G = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$\leftarrow P \quad \quad \quad \leftarrow I_k \quad \quad \quad$ (E-1720)

Step 2 : Find the parity check matrix H :

$$H = [I_{n-k} : P^T]$$

$$n = 6 \text{ and } k = 3 \therefore n - k = 3$$

$$\therefore H = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right]$$

$\leftarrow I_{n-k} \quad \quad \quad \leftarrow P^T \quad \quad \quad$ (E-1721)

Step 3 : Find the syndrome :

Received code word $r = 001110$.

$$\text{Syndrome } S = r \cdot H^T \quad \quad \quad \text{(E-1722)}$$

$$= [001110] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{H^T}$

Syndrome is same as the first row of H^T matrix

$$\therefore S = [100]$$

...Ans.

Step 4 : Obtain the correct code word :

Since the syndrome corresponds to the first row of the H^T matrix, the first bit in the received code word is erroneous.

$$\text{(E-1723)} \therefore \text{Corrected codeword} = \textcircled{1} 0 1 1 1 0$$

Corrected bit

...Ans.

Ex. 8.11.3 : The generator matrix for the (7, 4) block code is shown below,

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

- Find all code words and its minimum distance.
- If the received code word is {0 1 0 1 0 1 1} check for error and correct if any.

May 12, 08 Marks

Soln. :

Step 1 : Obtain the code words :

- The given code is (7, 4) code that means number of message bits will be 4. Thus the messages will be 0000, 0001, 1111.
- For message $m_0 m_1 m_2 m_3 = 0000$ the code word can be obtained as follows :

Message bits : $m_0 m_1 m_2 m_3 = 0000$

- From the given generator matrix the P matrix is as follows

$$P = P_{4 \times 3} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$[b_0 b_1 b_2] = [0, 0, 0, 0] \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = [000]$$

$\therefore$ Code word = 0000 000 **...Ans.**

- Similarly we can find the remaining code words.

Step 2 : Check and correct the error :

Given : Received code word = 0101011

Obtain the parity check matrix and its transpose :

$$H = [P^T : I_{n-k}]$$

- The transpose of P is given by

$$P^T = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\therefore H = \begin{bmatrix} 1 & 0 & 1 & 1 & : & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & : & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & : & 0 & 0 & 1 \end{bmatrix}_{3 \times 7} \quad \text{... Parity check matrix}$$

$$\therefore H^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{... Transpose of parity check matrix}$$

Decoding of the received code word :

- Received code word $Y = 0101011$.
- The syndrome of this code word is given by

$$S = YH^T = [0101011] \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = [101]$$

$$\therefore S = [1, 0, 1]$$

- This corresponds to the fourth row of H^T , Hence the error pattern is

$$E = 0 \ 0 \ 0 \ \textcircled{1} \ 0 \ 0 \ 0$$

Error (E-1740)

- And the correct code word is given by

$$X = E \oplus Y$$

$$X = [0001000] \oplus [0101011]$$

$$= [0 \ 1 \ 0 \ \textcircled{0} \ 0 \ 1 \ 1]$$

Corrected bit (E-1741)

Ex. 8.11.4 : Obtain code words for (6, 3) LBC which has generator matrix of $G = [100101; 010011; 001110]$. Find all possible code words. Obtain corrected code word, if received code word is $r = [001110]$. **Dec. 18, 8 Marks**

Soln. :

Part I : To obtain all the code words :

- Given generator matrix is,

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$I_{3 \times 3} \quad P_{3 \times 3} \quad \text{(E-2015)}$

- This is a (6, 3) LBC hence $k = 3$ (message bits)

1. Code word for 001 :

$$\text{Parity bits } [b_0, b_1, b_2] = [m_0, m_1, m_2] [P]$$

$$\therefore b_0, b_1, b_2 = [001] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = [110]$$

- $\therefore$ Code word for 001 is,

$$\text{Codeword} = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Message Parity (E-2016)

2. Other code words :

- Similarly we can obtain the code words for all other messages. They are listed in Table P. 8.11.4.

Table P. 8.11.4

Message word	Parity bits	Code word
0 0 0	0 0 0	0 0 0 0 0 0
0 0 1	1 1 0	0 0 1 1 1 0
0 1 0	0 1 1	0 1 0 0 1 1
0 1 1	1 0 1	0 1 1 1 0 1
1 0 0	1 0 1	1 0 0 1 0 1
1 0 1	0 1 1	1 0 1 0 1 1
1 1 0	1 1 0	1 1 0 1 1 0
1 1 1	0 0 0	1 1 1 0 0 0

Part II : Obtain the correct code word :

1. Obtain the parity check matrix :

$$H = [I_{n-k} : P^T]$$

$$\therefore H = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$\underbrace{\hspace{2cm}}_{I_{3 \times 3}} \quad \underbrace{\hspace{2cm}}_{P^T}$

(E-2017)

2. Find the syndrome :

Received code word $r = [001110]$

$$\text{Syndrome } S = r.H^T$$

(E-2018)

$$\therefore S = [001110] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$\underbrace{\hspace{2cm}}_{H^T}$

Syndrome is same as the fifth row of H^T matrix.

$$\therefore S = [011]$$

3. Obtain the correct code word :

Since the syndrome matches with the fifth row of H^T matrix, the fifth bit in the received code word is wrong.

$$\therefore \text{Correct codeword} = 0 \ 0 \ 1 \ 1 \ 0 \ 0$$

Corrected bit (E-2020)

Soln. :

Step 1 : Generator matrix (G) :

$$G = [I | P] = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$\underbrace{\hspace{2cm}}_{I_4} \quad \underbrace{\hspace{2cm}}_{P}$

(E-2073) Fig. P. 8.11.5(a)

Step 2 : Code vectors for the messages 1100, 0011 :

We know that, $B = MP$

$$1. \ [b_0, b_1, b_2] = [1100] \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} = 101$$

$$\therefore \text{Code word} = 1100101$$

...Ans.

$$2. \ [b_0, b_1, b_2] = [0011] \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} = 010$$

$$\therefore \text{Code word} = 0011010$$

...Ans.

Step 3 : Find the parity check matrix :

The parity check matrix $H = [P^T : I_{n-k}]$

$$\therefore H = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$\underbrace{\hspace{2cm}}_{P^T} \quad \underbrace{\hspace{2cm}}_{I_3}$

(E-2074) Fig. P. 8.11.5(b)

$$\therefore H^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 4 : Find the corrected codeword :

$$\text{Given } R = 0111101$$

$$\therefore \text{Syndrome } S = R.H^T$$

$$= [0111101] \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 100$$

Syndrome "S" corresponds to 5th row of H^T

(E-2075) Fig. P. 8.11.5(c)

This syndrome is equal to the fifth row of H^T .

Ex. 8.11.5 : For a systematic (7,4) LBC, the parity matrix is given by $[110; 011; 111; 101]$

1. Construct generator matrix.

2. Find code vectors for messages 1100, 0011.

3. If the received code vector is $R = 0111101$, find the corrected code word.

May 19, 7 Marks